ResearchForge

研究報告品質全面升級

Report Intelligence Upgrade - QA and Acceptance Record

Executive verdict
PARTIAL_REPORT_INTELLIGENCE_UPGRADE。核心 report-intelligence vertical slice 已完成並通過 configured regression；但 project_sources/ 與 13 份指定 DOCX/PDF 缺失，七題 post-slice live 全鏈驗收為 0/7，因此不能宣稱完整 PASS。

驗證面	結果	解讀
Configured suites	828 / 828 PASS	Report Factory 395；API Server 433
精確具名驗收	28 / 28 present	核心具名子集 123 / 123 PASS
A-G benchmark	7 / 7 PASS	Provider-free deterministic contracts；0 regression
Live formal cases	0 / 7 complete	不能以合成 benchmark 或 prior artifacts 代替
Latest actual model run	BLOCKED	Sentence claim/citation entailment 與 fallback gates 正確阻擋 export

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驗證原則：功能、資料與輸出必須分開證明；測試檔存在不等於 live product acceptance。

1. 問題、根因與升級邊界

- 舊流程的 Raw abstract、Evidence Card、section draft 與 writer 責任邊界不夠完整，容易形成摘要改寫、來源拼接與引用堆疊。
- Facts、claim graph、visual/export 已有良好基礎，但存取層級、全文語義段落、摘要/全文衝突、異質性、矛盾與統計解讀未集中成一份 canonical intelligence model。
- 數值能被抽出或畫圖，不代表已正確解釋；OR、相關、CI、p 值、橫斷研究與混雜因素需要獨立 deterministic rules。
- 既有品質檢查分散於 citation/export/render gates，缺少十面向總評、失敗位置、retry stage 與 retry cap。

本輪邊界
完成安全的 vertical slice，接入 canonical plan、writer、formal document 與 export integrity；沒有把缺少的 live 七題、未接通的 Image2 或不存在的 project_sources 包裝成已完成。

2. 新 canonical pipeline

階段	主要輸出	責任
Source resolution	canonical source + access tier	同來源去重、取得層級、failure truthfulness
Evidence resolution	semantic section + conflict ledger	Results 優先；Discussion 不冒充直接結果
Atomic Facts	nullable traceable facts	缺值保持 null；保留 source span/locator/verification
Report intelligence	knowledge, heterogeneity, stats, contradiction	概念與研究差異先於文字生成
Routing	AudienceProfile + ReportArchetype	控制深度、圖表、章節、最低證據條件
Narrative	teach-back + reconstruction	writer 不接 raw abstracts；citation identity 後接回
Verification	claim/citation/number/editor audits	逐 claim identity、錯誤來源、source echo 或 editor 未真正去重皆 fail closed
Quality/export	10 dimensions + retry/blockExport	canonical Preview/Markdown/DOCX/PDF

3. 資料模型與核心行為

3.1 Atomic Evidence Fact

Schema 現在包含 sourceType、evidenceLevel、sourceAccessLevel、sectionType、studySetting、pValue、moderator、confounders、authorInterpretation、authorLimitations、extractionConfidence 與 verificationStatus；來源未提供的欄位維持 null。

3.2 Full-text / abstract resolution

能力	實作結果
六級存取	complete full text、structured full text、verifiable excerpt、complete abstract、partial abstract、bibliographic only
語義段落	背景、問題、樣本、eligibility、design、exposure、outcome、measurement、statistics、results、sensitivity、discussion、mechanism、limitations、conclusion
衝突	保留 abstract/full-text conflict；優先 Results full text；confidence penalty 0.25
Fail closed	無 locator 數值不可 verified；extraction failure 不產生 ghost evidence

3.3 Knowledge, heterogeneity and contradiction

- Knowledge nodes 覆蓋 population、exposure、outcome、mediator、moderator、confounder、setting、method、direction、applicability 與 gap。
- Knowledge edges 保留 supporting/counter sources、directness、statistical relation vs author inference、causal eligibility 與 uncertainty。
- Comparability 僅允許 COMPARABLE、PARTIALLY_COMPARABLE、DIRECTION_ONLY、NOT_COMPARABLE；OR/RR/HR 不互相 pooling。
- TikTok、短影音、短影音成癮、一般社群媒體；LST、氣溫、熱舒適、UHI；GVI、canopy、NDVI、green coverage 維持概念隔離。
- Contradiction 保留反方向、null/mixed 結果、單一來源與替代解釋，不為了流暢隱藏反例。

4. 統計、視覺與讀者路由

4.1 Statistical Interpretation Layer

家族	核心保護
OR / RR / HR	保留 ratio 與 null=1；OR 不改寫為風險增加百分比
Correlation	方向與關聯強度；明確禁止因果
CI / p value / null	CI crossing null 不宣稱穩定；p > .05 不等於沒有關係
Mean / median / SD / IQR	不同量表與計分方向不可直接比較
Regression / standardized beta	保留單位、模型與 adjustment status
Mediation / interaction	模型路徑不冒充已證明機制
Model fit / non-significant	解釋樣本內變異；不等於預測或因果有效

4.2 Visual semantics and Image2

- 比例 -> proportion bar；OR/RR/HR + CI -> forest plot；correlation -> matrix；mediation/interaction -> path diagram。
- 所有研究數字、軸、CI 與 label 由 structured facts deterministic 生成並有 datum provenance。
- Image2 prompt 若含研究數字、百分比、OR/RR/HR/CI 即不 truthful；目前沒有 live Image2 provider，使用 deterministic fallback。

4.3 Audience and archetype

Router	支援	不是只改字數
AudienceProfile	一般、高中、大學、專業、決策	術語、句長、統計、方法、圖表、引用、限制、建議、證據分級
ReportArchetype	綜述、比較、機制、實驗、技術、政策、市場、爭議、有限證據、證據地圖	各自 blueprint、evidence/visual requirements、comparison method、conclusion form、minimum conditions

5. Writer、editor、verifier 與 QualityGate

5.1 Writer-safe contract

- Teach-back packet 去身分化，不含 sourceSpan、rawText、supportSpan、abstract 或 snippet values。
- Academic reconstruction 取得 normalized facts、claim graph、stat interpretations、heterogeneity、contradiction、audience/archetype 與 citation identity。
- 實際 prompt sentinel tests 證明 RAW_ABSTRACT/RAW_SUPPORT_SPAN/RAW_NUMERIC_SENTENCE 不會進入 writer。

5.2 Whole-document editor

Editor 可合併重複、修轉場、統一術語與順序，但不能改 citation identity/order、number、year、location、method、effect measure 或 evidence boundary；post-edit 再跑 sentence claim/citation entailment、source-echo 與 edit 前後重複度 checks。若 exact/near-duplicate burden 沒有下降，不得標為 applied。

5.3 Ten-dimension ReportQualityGate

Dimension	Deterministic evidence	Retry target
Evidence integrity	source/evidence/locator traceability	evidence extraction
Citation precision	entailment + identity	claim/citation mapping
Synthesis	source-order/stitching	narrative planning
Statistical correctness	number facts + interpretation	evidence extraction
Coherence	transition + repetition	editor / narrative
Visual effectiveness	type + provenance + render	visual router
Directness honesty	overclaim + sufficiency	readiness/claim strength
Contradiction	material counterevidence	narrative planning
Audience fit	profile + depth + archetype	audience/report routing
Document quality	canonical parity + CJK	canonical/exporter fonts

Retry policy
Default maxRetries=2。超過上限保留 audit plan，並輸出 Limited Evidence Brief 或 QA failed；目前 retry instructions 已完成，但 upstream orchestration loop 尚未完整自動化。

6. Test evidence

Suite	PASS	FAIL	Status
Report Factory configured	395	0	PASS
API Server configured	433	0	PASS
Total	828	0	PASS
Focused named subsets	123	0	PASS
TypeScript typecheck	1	0	PASS
git diff --check	1	0	PASS

6.1 Exact named acceptance

Named test	Named test
writer_does_not_receive_raw_abstracts	full_text_sections_are_not_confused
atomic_facts_are_source_traceable	concepts_are_not_falsely_merged
claim_graph_contains_support_and_counterevidence	heterogeneity_classification_is_consistent
no_cross_study_false_comparison	statistical_interpretation_matches_type
odds_ratio_not_rendered_as_percentage	correlation_not_claimed_as_causation
all_chart_values_are_traceable	visual_type_matches_statistic
main_body_contains_result_visuals	image2_does_not_generate_research_numbers
no_source_order_narrative	no_paper_by_paper_stitching
no_blanket_citation_clusters	whole_document_editor_removes_repetition
audience_profiles_change_explanation_depth	report_archetype_matches_question
contradictory_evidence_is_not_hidden	quality_gate_routes_failures_correctly
limited_evidence_downgrade_is_honest	claim_citation_reference_identity
docx_pdf_preview_content_parity	cjk_render_preserved
body_paragraph_first_line_indent	benchmark_suite_has_no_unexplained_regression

7. A-G regression matrix

ID	Scenario	Checks	Status	Diagnostics
A	多筆定量研究方向一致	統計抽取；視覺路由；跨研究可比性	PASS	interpretations=3；comparisons=3
B	有數字但統計指標不可直接比較	不錯誤合併；不建立虛假排名；保留方向性綜合	PASS	status=DIRECTION_ONLY；pooling=false
C	研究結果互相矛盾	反例保留；無顯著結果保留；替代解釋	PASS	status=contradictory；counter=1；null=1
D	只有少量摘要證據	誠實降級；不虛構全文；不硬填章節	PASS	archetype=limited_evidence_report；downgradedFrom=literature_review
E	不同語言來源	概念標準化；引用身分一致；不重複計數	PASS	status=COMPARABLE；canonicalSources=1
F	沒有直接證據	背景不冒充直接結果；顯示研究缺口	PASS	evidenceLevel=background；archetype=limited_evidence_report
G	方法型或實驗設計問題	正確 archetype；變項與假設；流程與驗證	PASS	archetype=experimental_design；sections=研究假設 | 變項操作化 | 樣本與分派 | 實驗流程 | 分析與驗證 | 風險與限制

Interpretation

A-G 7/7 PASS、unexplained regressions=0。這些是 fixed provider-free contracts，用於防止 deterministic intelligence regression；不得替代 live search、writer、render 與四格式驗收。

8. Before / After

面向	Before	After
Writer	source/evidence material boundary 不完整	facts/claims/intelligence safe projection
Fusion	易沿來源或卡片序列	concept-first graph + stitching/source-order gates
Statistics	抽取與呈現為主	independent family/CI/p/causal/comparability rules
Contradiction	散落在 cards/prose	counter/null/mixed + alternatives + quality gate
Audience	輸入與篇幅導向	5 profiles 控制多個解釋維度
Archetype	較固定骨架	10 blueprints + minimum conditions
Quality	分散 audits	10 dimensions + locations + retry + blockExport

9. Live acceptance truth table

Case	Required	Status	Evidence
short_video_sleep	比例、OR、相關與中介路徑	BLOCKED	Latest actual Ollama run: provider=ollama, model=qwen2.5:3b, reportGenerationStatus=failed, exportSuppressed=true.
urban_heat	跨場域、跨指標與近直接證據	BLOCKED	No post-slice live rerun.
contradictory_topic	結果互相矛盾	BLOCKED	Deterministic Benchmark C passed; no live report bundle was generated.
full_text_sufficient	足夠全文證據	BLOCKED	No supplied full-text acceptance corpus and no live report bundle.
abstract_only	只有摘要的有限證據	BLOCKED	Deterministic Benchmark D passed; no post-slice live report bundle.
no_direct_evidence	幾乎沒有直接證據	BLOCKED	Deterministic Benchmark F passed; no post-slice live report bundle.
non_literature_review	實驗設計或技術比較	BLOCKED	Deterministic Benchmark G passed; no post-slice live report bundle.

Why this is PARTIAL

PASS requires at least one complete live Source -> Evidence -> Facts -> Knowledge -> Heterogeneity -> Statistics -> Visual -> Narrative -> Editor -> Quality -> Preview/Markdown/DOCX/PDF path. This post-slice evidence does not contain one. Prior accepted bundles are baseline only.

9.1 Prior accepted artifacts

Artifact	Formats	Caveat
urban_heat	Preview / MD / DOCX / PDF	Prior live baseline; predates this slice
short_video_sleep	Preview / MD / DOCX / PDF	Prior live baseline; predates this slice
latest actual short_video	Preview/audit only	Ollama qwen2.5:3b failed post-edit entailment; export suppressed

10. Files and traceability

Location	Role
lib/api-zod/src/research-grounding.ts	Canonical schemas
api-server/.../evidenceResolution.ts	Full-text/abstract access, sections, conflict, locator
api-server/.../researchGroundingPipeline.ts	Resolver integration and atomic metadata
report-factory/.../reportComposer.ts	Extended Atomic Facts and claim graph
report-factory/.../reportIntelligence.ts	Knowledge, heterogeneity, statistics, contradiction, routing
report-factory/.../structuredReportPlan.ts	Canonical intelligence plan
api-server/.../report-writer.ts	Writer-safe packets and scoped intelligence
report-factory/.../reportQualityGate.ts	Ten-dimension quality and retry
report-factory/.../formalReportDocument.ts	Quality/blockExport integration
report-factory/.../reportIntelligenceBenchmark.ts	Fixed A-G suite

11. Remaining work

- project_sources/ 以及 7 DOCX + 6 PDF 未提供；無法讀取其方法學 rubric，也未將其當學術證據。
- 七個正式驗收題尚未完成 post-slice current-architecture live rerun。
- Source ingestion 以 HTML/PDF text/GROBID/OCR 為主；DOCX/EPUB 尚無原生 adapter。
- 複雜多層模型、生存曲線、Bayesian posterior、network meta-analysis 尚無專用 statistical family。
- 十種 archetype 只有 deterministic router/contract coverage，尚未各自完成 live writer/export acceptance。
- Quality retry instructions 與 cap 已完成；自動回到每個 upstream stage 的完整 orchestration loop 尚未完成。
- 中央 ModelCapability/Context/Cost registry 尚未建立；既有 adapter 只有 purpose/model/timeout/token caps。
- Live Image2 provider 未接通；目前只允許 truthful deterministic fallback。

Final token
PARTIAL_REPORT_INTELLIGENCE_UPGRADE

This QA PDF is the only new PDF produced by this slice. Prior research-report PDFs are linked as historical evidence and are not re-labeled as current outputs.